

BUFFALO NARROWS, SK, Canada

WMO: 710770

Lat: 55.8409N

Lon: 108.4208W

Elev: 1444

StdP: 13.95

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.6	(c) -26.4	(d) -38.4	(e) 0.6	(f) -31.1	(g) -33.4	(h) 0.9	(i) -26.0	(j) 23.2	(k) 8.1	(l) 20.5	(m) 8.9	(n) 4.0	(o) 300	(p) 0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.2	(c) 81.1	(d) 63.6	(e) 77.8	(f) 62.1	(g) 74.9	(h) 60.9	(i) 66.0	(j) 76.6	(k) 64.4	(l) 74.2	(m) 62.9	(n) 71.9	(o) 8.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	88.4	69.7	60.7	83.7	68.1	59.3	79.5	66.6	31.6	76.8	30.3	74.1	29.2	72.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 18.2	(c) 15.9	(e) -39.6	(f) 87.4	(g) 5.8	(h) 2.7	(i) -43.7	(j) 89.4	(k) -47.1	(l) 91.0	(m) -50.3	(n) 92.5	(o) -54.5	(p) 94.5		
(a) 20.9	(b) 18.2	(c) 15.9	(e) -39.6	(f) 87.4	(g) 5.8	(h) 2.7	(i) -43.7	(j) 89.4	(k) -47.2	(l) 91.0	(m) -50.5	(n) 92.5	(o) -54.7	(p) 94.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	1.4	6.2	18.8	35.3	48.8	59.3	64.3	61.9	51.8	37.8	19.6	5.8		
	DBStd	24.99	16.90	14.60	15.37	10.69	8.83	6.21	4.80	5.93	7.59	8.50	11.98	15.52		
	HDD50	7030	1507	1228	968	449	131	7	0	1	68	387	912	1372		
	HDD65	11280	1972	1648	1433	891	506	187	71	131	398	844	1362	1837		
	CDD50	1331	0	0	0	8	93	286	443	369	123	9	0	0		
	CDD65	103	0	0	0	0	2	17	48	34	2	0	0	0		
	CDH74	830	0	0	0	2	42	163	347	244	32	0	0	0		
	CDH80	147	0	0	0	0	6	31	60	46	4	0	0	0		
	WSAvg	7.2	6.7	6.8	7.2	7.5	7.4	6.9	6.7	6.9	7.7	8.1	7.5	6.8		
	PrecAvg	18.1	1.0	0.7	0.7	1.1	1.4	2.7	3.1	2.4	1.9	1.2	0.9	0.9		
(15) Precipitation	PrecMax	25.0	4.1	1.9	1.4	2.6	2.9	8.0	6.1	6.9	4.8	3.1	2.6	2.0		
	PrecMin	10.3	0.3	0.0	0.0	0.2	0.0	0.8	1.1	0.1	0.6	0.0	0.3	0.2		
	PrecStd	3.5	0.7	0.5	0.4	0.6	0.7	1.5	1.3	1.4	0.8	0.7	0.6	0.4		
	PrecStd	3.5	0.7	0.5	0.4	0.6	0.7	1.5	1.3	1.4	0.8	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	43.9	52.1	69.7	79.3	84.0	85.3	85.0	78.2	66.4	46.6	39.7		
		MCWB	36.4	36.7	40.8	49.9	57.6	62.8	66.4	65.2	62.0	53.6	39.7	34.8		
	2%	DB	35.1	35.9	46.8	61.6	73.1	78.2	81.4	79.8	72.4	58.3	41.3	34.1		
		MCWB	32.1	31.1	37.8	46.6	54.9	60.2	64.7	64.1	59.1	48.6	36.6	31.4		
	5%	DB	30.2	31.0	42.2	56.2	68.8	74.8	77.9	76.3	67.7	53.6	37.4	30.4		
		MCWB	28.3	28.2	35.4	43.7	52.8	58.7	63.0	62.7	56.7	46.3	33.7	28.7		
	10%	DB	25.3	26.9	38.3	51.4	64.3	71.3	74.8	73.0	63.8	49.5	34.2	26.0		
		MCWB	23.9	25.0	33.2	41.2	51.0	57.4	62.0	61.4	54.9	43.5	32.0	24.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	36.9	41.4	51.3	60.1	65.6	68.8	68.0	63.6	54.4	40.4	35.7		
		MCDB	40.8	43.1	50.6	67.6	72.6	78.3	80.0	80.2	74.0	65.4	45.6	38.9		
	2%	WB	32.2	31.7	38.2	47.0	57.1	63.0	66.6	65.6	60.4	50.0	37.1	31.5		
		MCDB	34.8	35.4	45.6	60.1	69.6	72.9	76.8	76.2	69.5	56.5	40.9	33.5		
	5%	WB	28.3	28.3	35.7	44.3	54.4	61.1	65.0	64.0	57.9	46.6	34.4	28.4		
		MCDB	30.1	31.0	41.4	55.2	65.7	69.8	74.7	73.5	65.8	52.9	36.8	30.4		
	10%	WB	23.8	25.0	33.4	41.4	52.0	59.3	63.4	62.5	55.7	44.1	32.1	24.6		
		MCDB	25.4	26.9	37.9	50.4	62.8	67.6	72.5	70.9	62.6	49.1	34.2	25.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	18.3	20.1	19.3	20.1	17.5	17.2	16.3	15.3	12.2	11.1	14.1		
		MCDBR	19.6	21.0	22.0	26.3	26.7	23.7	22.9	22.1	22.5	19.7	13.6	17.3		
		MCWBR	17.0	17.4	14.8	14.9	12.8	10.2	10.0	10.1	12.0	11.5	9.9	15.5		
	5% WB	MCDBR	19.4	20.1	20.8	25.2	23.2	18.8	19.8	19.9	20.3	18.6	12.5	17.2		
		MCWBR	17.1	16.8	14.3	14.7	12.0	9.3	9.7	9.9	11.7	11.5	9.7	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.261	0.271	0.315	0.331	0.357	0.368	0.348	0.310	0.284	0.238	0.216		
	taud		2.318	2.309	2.410	2.459	2.476	2.420	2.411	2.459	2.563	2.573	2.427	2.282		
	Ebn at Noon		231	263	290	289	289	281	275	274	273	256	229	208		
	Edh at Noon		15	23	27	30	32	34	34	30	23	18	14	12		
(44) All-Sky Solar Radiation	RadAvg		239	558	1060	1509	1771	1774	1739	1438	982	532	230	156		
	RadStd		23	33	65	105	177	148	107	122	110	81	30	14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page